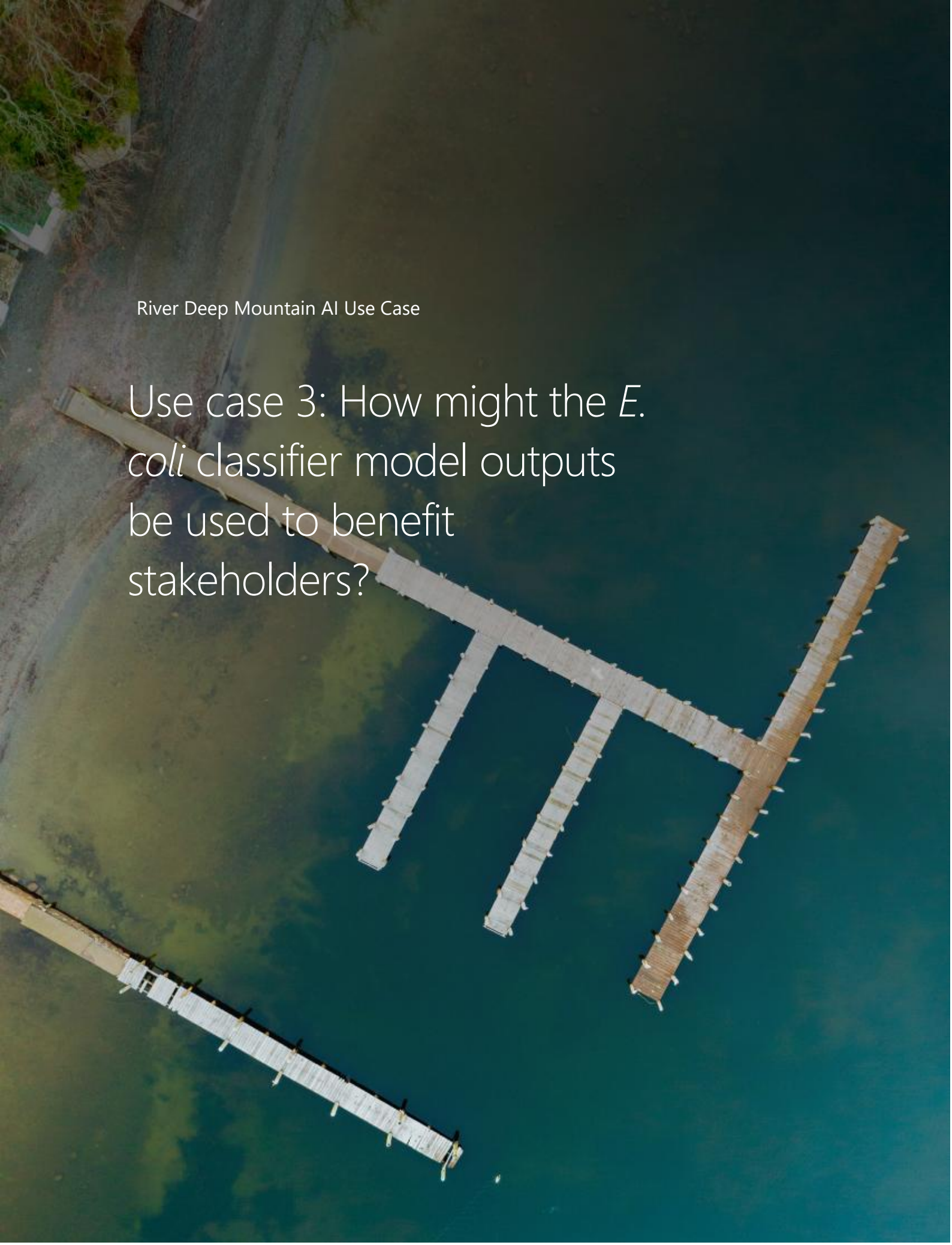

River Deep Mountain AI Use Case

Use case 3: How might the *E. coli* classifier model outputs be used to benefit stakeholders?

River Deep Mountain AI

Version overview

Version	Date	Authors	Comments
0.1 Draft	January 2026	Daisy Allen (WRc), Nancy Battersby (WRc), Kata Farkas (WRc), Giles Exley (WRc)	Comments from WRc, NWG, ADAS, Cognizant and the Rivers Trust addressed.
0.2 Draft	February 2026	Daisy Allen (WRc), Nancy Battersby (WRc), Kata Farkas (WRc), Giles Exley (WRc)	Comments from WRc, NWG, ADAS, Cognizant and the Rivers Trust addressed.
1.0 Final Draft	February 2026	Daisy Allen (WRc), Nancy Battersby (WRc), Kata Farkas (WRc), Giles Exley (WRc)	Final version of report.
2.0 Final	March 2026	Daisy Allen (WRc), Nancy Battersby (WRc), Kata Farkas (WRc), Giles Exley (WRc)	Links and disclaimer updated. Watermarks removed. Removed author names.

Approval

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Executive Summary

This report explores how the RDMAI Open (open source) predictive model for *Escherichia coli* (*E. coli)* could be applied across multiple water-related use cases to support public health, regulatory compliance, and operational decision-making.

The RDMAI classifier model has the potential to give real-time or near real-time predictions of higher risk and lower risk *E. coli* levels in riverine and coastal waters, providing timely warning of faecal contamination risks. Two versions of the model exist: a 'light' model using land use, weather, and remote sensing data, and an 'advanced' model incorporating additional water quality parameters. Both aim to deliver actionable insights beyond traditional laboratory sampling, which is limited by cost, timing, and geographic coverage.

The document outlines three major application areas:

1. **Predicting Bathing Water Quality**

   The RDMAI classifier model could be used to inform recreational water users and water industry stakeholders by forecasting *E. coli* levels aligned with relevant UK regulatory thresholds. This enables proactive risk communication and management, reducing exposure to contamination during high-risk events. Existing case studies, such as SEPA's bathing water forecast and Wessex Water's Warleigh Weir model, demonstrate practical workflows for public alerts and integration with digital platforms.

2. **Supporting Shellfish Harvesting Risk Management**

   The RDMAI classifier model could be used to classify shellfish beds according to hygiene standards set by the Food Standards Agency, helping farmers and regulators anticipate contamination events and optimise harvesting schedules. Case studies in this report demonstrate other models that have achieved high accuracy in predicting *E. coli* classifications in estuarine harvesting locations, offering considerable value for operational planning and consumer safety.

3. **Catchment Risk Assessment for Drinking Water Abstraction**

   The RDMAI *E. coli* classifier model could be used to complement statutory monitoring by forecasting microbiological risks at abstraction points, particularly during rainfall-driven contamination events. Case studies in this report demonstrate other models that can predict *E. coli* peaks in advance of laboratory testing results. Such insights could enable water suppliers to implement timely control measures and integrate predictive insights into water safety planning frameworks.

The applications above would require the use of the RDMAI models for forecasting. While the RDMAI light classifier model can provide real-time estimations of higher risk and lower risk *E. coli* levels, the inputs to the RDMAI light model, weather data

and MODIS remote sensing data, can be forecast. Therefore, there is potential to use the RDMAI light model to forecast future *E. coli* levels.

Across all potential use cases, the RDMAI model offers benefits such as improved spatial and temporal resolution, early warning capabilities, and potential integration with existing regulatory and operational systems. However, limitations include data availability, model accuracy, and the need for calibration to local conditions. Predictive outputs should be treated as indicative rather than definitive and cannot replace statutory monitoring requirements.

The report concludes by identifying opportunities for scaling, integration with policy, and refinement of the RDMAI model to include additional data sources (e.g., shellfish-derived *E. coli* data). These steps could enhance its utility for public health protection, industry compliance, and sustainable water resource management.

Introduction

The primary output of the RDMAI Open (open source) classifier model is to predict higher risk and lower risk *Escherichia coli* levels in riverine and coastal waters in Great Britain. Because *E. coli* originates from the gastro-intestinal tract of humans and warm-blooded animals, its presence indicates faecal contamination and potential presence of enteric pathogens (Ashbolt et al., 2001). High *E. coli* levels are linked to an increased risk of gastrointestinal illness in people exposed to contaminated water (Cabelli et al., 1982; Kay et al., 1994; Medema et al., 1997; O'Conner et al., 2002).

There are two versions of the RDMAI classifier model, the 'light' model and the 'advanced' model, which require different data inputs.

- The **light** version requires land use, weather, and MODIS remote sensing data.
- The **advanced** version requires land use, weather, and MODIS remote sensing data, and additionally, water quality measures including water temperature, suspended solids, ammoniacal, nitrate, nitrite, and phosphate.

To date, the light model has been trained on data with geographic coverage of Great Britain, whereas the advanced model has been trained on data with geographic coverage of England. The accuracy of the model in a particular catchment will depend on the temporal and spatial resolution of the data used to train the algorithms.

Various algorithms were investigated during the development of the RDMAI *E. coli* classifier models. These included:

- K-Nearest Neighbours (KNN): Classifies data points based on the majority class of their closest neighbours.
- Support Vector Classification (SVC): finds the best boundary (hyperplane) that separates different classes, performing well for binary classification tasks.
- XGBClassifier: Applies XGBoost for classification, effectively handling complex patterns in *E. coli* presence.

The XGBoost classifier was found overall to be the most accurate when applied to data for England and Great Britain.

Shortcomings in RDMAI model performance against measured data were demonstrated in the model validation. The model shows a bias towards the majority class, resulting in a relatively high rate of false negatives (FNs) (15% and 27% for the best performing advanced and light models, respectively during validation). In addition, the classifier is sensitive to the chosen threshold, which affects the FN/FP balance. Benchmarking of the RDMAI *E. coli* models also showed that they have relatively limited accuracy at this stage. The bias of the models to the majority class (low *E. coli* measures) results in under simulation of high *E. coli* concentrations, and a high rate of false negatives. The model performs better when applied nationally than to a specific site. The model false negatives when applied to a specific site were between 37% and 66%, which improved to between 19% and 26% when applied

nationally. Reports on the model performance (benchmarking and validation) can be found here.

While the RDMAI *E. coli* models in their current state are not sufficiently accurate to replace current water company and EA site-specific models, there is potential for their application alongside citizen science data and site-specific models. Since the RDMAI *E. coli* models are open source, there is a potential for further improvement of model performance, by for example training them for a specific site.

The RDMAI light models can be used to estimate *E. coli* levels and risk historically, and near-real time. However, since the inputs to the light models, namely weather and MODIS remote sensing data, can be forecast, there is potential to use the RDMAI light models in a forecast capacity. The potential use cases discussed in this report assume the use of the RDMAI light models as forecasting models.

There are many potential applications for the RDMAI light model as a forecasting model, which are explored herein as 'use cases'. Although the model to-date has been developed with the purpose of estimating bathing water quality, there are further opportunities for extending its application. The objective of this report is to explore these potential use cases with example workflows (inputs, outputs and end-user interface), value propositions, limitations, integration and/or scaling.

Predicting Bathing Water Quality

In recreational waters, people may ingest or come into direct contact with water contaminated by faecal pollution. Regulatory frameworks around the world rely on *E. coli* as a primary indicator to assess water quality, protect public health and guide the management of beaches, rivers and lakes used for bathing and other leisure activities.

The RDMAI models aim to inform recreational water users of health risk in riverine and coastal waters in Great Britain. Used as forecast models, the RDMAI models have the potential to inform, ahead of time, changes in *E. coli* levels.

The output of the model provides information relevant to stakeholders within the water industry to provide actionable risk-management decisions, and to citizens who use waterbodies for recreational purposes. The classification model in its current form provides binary outputs: 'lower risk or 'higher risk', based on thresholds of *E. coli* concentration associated with risk of illness in bathing waters. Threshold 1 aligns with national and international guidelines from the UK, US Environmental Protection Agency (EPA), and the World Health Organisation (WHO). Threshold 2 is based on the EA guidelines for *E. coli* for a good classification in coastal waters.

- Threshold 1: E. coli > 1000 CFU/100mL = 'Higher Risk'; E. coli < 1000 CFU/100 mL = 'Lower Risk'

- Threshold 2: E. coli > 500 CFU/100mL = 'Higher Risk'; E. coli < 500 CFU/100 mL = 'Lower Risk'

The RDMAI classifier model can be run with either of the above two thresholds.

To-date, bathing waters have principally been assessed using laboratory analysis of samples, which is restricted by both operational (who, when and where) and cost factors. In addition, sample results only provide data for past events, where pollution may have already occurred, and the public have been exposed to a risk event. The opportunity for the RDMAI classifier model is to provide actionable insights, ahead of time even in waterbodies where sampling is limited.

This report provides use cases of existing models that predict bathing water quality; these include Wessex Water: Warleigh Weir and the SEPA Bathing Water Forecast. The RDMAI model can adopt similar workflows and provide potential for integration and/or scaling.

Case Study 1: SEPA Bathing Water Forecast (SEPA, n.d)

Summary

The Scottish Environmental Protection Agency (SEPA) provide an electronic message system which communicates to the public a forecast of predicted bathing water quality. The authority recognises that waters may be at risk of not meeting European standards (Bathing Water Directive 2006/7/EC) during or after rainfall events. The signage network communicates precautionary forecasts via the website and text message and will let the end-user know if the 'water quality is forecast to be acceptable today' or to advise otherwise.

The forecasts are made based on rainfall and the hydrogeological information network and the purpose of the signage is to communicate the model prediction. The model covers Scottish bathing water sites only during the bathing season. In addition, the predictions do not capture any overflows (storm overflows, settled storm sewage overflows and emergency overflows); however, a separate interactive real-time overflow map has been provided for this purpose (Scottish Water, n.d.).

The SEPA Bathing Water Forecast does not predict *E. coli* risk levels, unlike the RDMAI classifier model. Therefore, the outputs cannot be interpreted within a regulatory context. There is potential for integration of the RDMAI model to provide more informed assessments of public health risk, by enabling the existing SEPA Bathing Water Forecast system to leverage *E. coli* risk-based predictions.

In addition, the example use case shows how bathing water forecasts can be shared with the public via an electronic messaging system and website, thereby reducing bathing during high-risk events, improving public-health and providing transparent communication to recreational water users.

SEPA Bathing Water Forecast (SEPA, n.d) Workflow:

Model Link: https://bathingwaters.sepa.org.uk/predictions/
INPUTS
Hydrogeological and rainfall data • Rain gauge measurements (rainfall intensity/volume) • River flow/river level measurements at gauge stations Optional/additional sources – for advanced models • Other environmental/categorical variables (e.g., tide times, sunshine/humidity, land-use / catchment pollution sources) • Historical data linking hydrology and rainfall to past water quality outcomes (indicator bacteria levels, event history)
PROCESSING/MODEL
For legacy model (site-specific criteria spreadsheet) (McPhail and Stidson, 2009): • Compare current rainfall and river flow values to historically defined thresholds for that site (derived from past data) For newer / advanced models (decision-tree / machine-learning based): • A decision tree (or ensemble of trees / boosted model) uses input variables (numeric and categorical) and applies weightings to optimise predictions.
OUTPUT
Daily forecast for each bathing site (e.g., 'Water quality is forecast to be acceptable today' or warning of less acceptable quality)
END-USER INTERFACE / COMMUNICATION
Electronic message signs at selected bathing water sites updated daily during bathing season – Picture 1. Online publication on SEPA website, accessible to public, updated daily. Phone / text-message service, local-authority / partner advisories for pollution events.

Picture 1: Electronic signage at bathing site. (https://www.scrimsign.co.uk/case-study/SEPA-Bathing-Water-Signs/)

Case Study 2: Wessex Water: Warleigh Weir *E. coli* prediction (Water Quality Live, n.d.)

Summary

Wessex Water has developed a machine learning based model for *E. coli* prediction at Warleigh Weir. The site is a popular bathing spot along the River Avon in Bath, England. The output of the model is communicated to the public via a website as a traffic-light style (orange/green) swim alert with text advising if there is indicated higher risk to bather's health from poor water quality. Swim alerts are set for when *E. coli* exceed 900 CFU/100mL and intestinal enterococci exceed 330 CFU/100mL, which match the classification for 'sufficient' in The Bathing Water Regulations (2013).

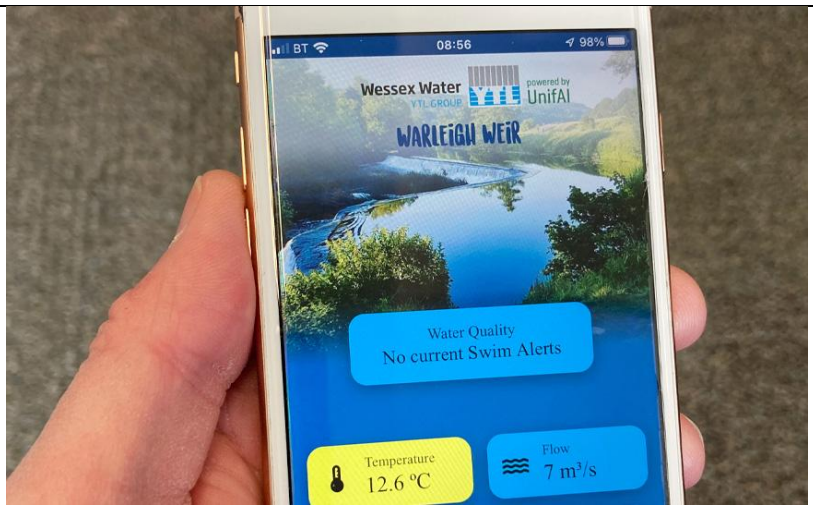
The half-hourly predictions are made based on rainfall, flow and water quality data specific to the Warleigh Weir catchment including ammonium, ammonia, chloride, conductivity, dissolved oxygen, pH, suspended solids, temperature and turbidity. The data is provided using *in-situ* sensors. The same machine learning algorithm is currently being trained for bathing water sites across Somerset, Wiltshire, Dorset and South Gloucestershire. The alert system is provided year-round, and provides measures of temperature, flow and rainfall in acknowledgment that other factors may contribute to interpretation of 'safe' waters for the purpose of recreation. Validation sampling is routinely undertaken to train the algorithms.

The example use case is specific to Warleigh Weir, and as real-time data from sensors is used to predict *E. coli* levels, scaling-up of the model would require the instalment of sensors at the location of interest to obtain water quality data. However, the RDMAI classifier model offers increased geographic coverage and could theoretically be applied to any location, with accuracy dependent on the quality and spatial distribution of training data.

The example use case shows how *in-situ* sensors and AI can be used to predict indicator bacteria levels and issue public-health alerts via a website, providing transparent communication between water company and recreational water users.

Warleigh Weir *E. coli* prediction (Water Quality Live, n.d.) Workflow:

Model Link: https://www.waterqualitylive.info/
INPUTS
Real-time sensor data (at Warleigh Weir) • Water temperature • Water level/flow • Rainfall Historical data to train model (Warleigh Weir at time of writing, with model training planned at bathing water sites across Somerset, Wiltshire, Dorset and South Gloucestershire) • Laboratory sample results- Indicator organisms (*E. coli* and intestinal *Enterococci*) • Sensor results- Including ammonium, ammonia, chloride, conductivity, dissolved oxygen, pH, suspended solids, temperature and turbidity
PROCESSING/MODEL
Uses real-time and derived variables from *in-situ* sensors at Warleigh Weir Applies machine learning algorithms trained on historical data
OUTPUT
Predicts real-time indicator organism contamination levels at the site Provides real-time data on other relevant parameters (temperature, flow and rainfall)
END-USER INTERFACE / COMMUNICATION
Online publication on website, accessible to public, updated daily – Picture 2. Traffic-light style system for end-users to inform recreational water users of indicative health risk. Additional information provided on temperature, flow and rainfall provided in acknowledgment of other factors that contribute to bathing safety. Statements regarding liability are readily available on the website, which states that it is the responsibility of the user to consider whether the water is safe to enter.



Picture 2: Website-based swim alert system. (https://www.wessexwater.co.uk/news/ai-technology-used-for-warleigh-weir-bathing-water-study)

Value Propositions

The RDMAI classifier model, used as a forecast model, could offer benefits for bathing water safety by enabling early warning of contamination risk before exposure occurs. Its thresholds align with various national and international guidelines, ensuring outputs are relevant for compliance and regulatory decision-making. The RDMAI model also can be applied to more locations than the Wessex Water use case, which is restricted to a specific bathing water site.

Unlike traditional laboratory sampling, which is limited by cost and timing, the RDMAI model delivers improved spatial and temporal resolution, enabling more frequent predictions, even for sites with minimal monitoring. This capability could support operational efficiency for water industry stakeholders by informing timely interventions such as signage updates and public alerts. Additionally, these interventions can be integrated with digital platforms including websites, mobile notifications, and electronic signage. These types of measures could improve transparency and trust with recreational water users.

Beyond public health benefits, the model also provides opportunities for benchmarking and validation against other AI/ML systems, helping to identify previously unrecognised risk areas and strengthen confidence in water quality management strategies

Limitations

Predictive models such as the RDMAI classifier can provide valuable insight, but they cannot offer comprehensive assessments of health risk. The provided case studies emphasise that forecasts are indicative only, and responsibility ultimately remains with the end user when deciding whether to enter the water. Traffic-light or risk-rating systems must clearly state that other factors, beyond predicted indicator-

bacteria levels, may influence actual water quality and health outcomes. Users of the RDMAI classifier will need to consider this when deploying the model.

A major limitation of predictive modelling is the potential for false negatives, where a model output indicates *E. coli* levels are below threshold when actual *E. coli* levels are above the threshold. This presents public-health and reputational risk. Conversely, false positives may trigger unnecessary warnings, reducing site use, harming local economies and diminishing public trust. These inaccuracies can also raise equity concerns, as misclassification may disproportionately affect certain segments of the population, such as wild swimmers. However, the RDMAI classifier model provides the user opportunity to train the model for a specific catchment (and as the model is open source, they could amend and adapt the code to suit their specific requirement) which could enable continuous improvement to the accuracy of the model's predictions.

Predictive systems must also operate within regulatory constraints i.e., they cannot replace statutory bathing-water monitoring required under legislation. Data availability presents further challenges; *E. coli* samples are typically collected weekly or fortnightly, providing a limited evidence base for training effective models, particularly during short-term pollution events or high-flow conditions.

Finally, practical considerations, such as ongoing maintenance, calibration and validation of sensors and data streams, impose significant operational demands and impact the cost-benefit of predictive models.

Next steps and opportunities

There is opportunity for the RDMAI classifier model to be integrated with existing use case models or used as an alternative. The obvious end-use in a bathing water scenario would be the prediction of potential health risk (based on *E. coli* thresholds) to recreational water users or other stakeholder groups. There is limited information on how these models could be used within individual water companies or shared between them. However, it is reasonable to assume that the RDMAI classifier model could help improve understanding of bathing water quality and determine whether specific sites are at risk of meeting compliance requirements.

The RDMAI classifier model could be combined with other existing models to expand geographic coverage and generate *E. coli* risk-based predictions for systems that do not currently produce this output. It could also be used to benchmark and validate other AI/ML models, helping to identify previously unrecognised risk areas by incorporating additional or different input variables.

There is also potential to support ongoing model training and validation using *in-situ* sensors (similar to the Warleigh Weir prediction model) or drone-collected data. These inputs may enhance spatial resolution and provide new parameters (e.g., algae proxies) to improve predictive accuracy. The model could also be used to guide risk-based deployment of sampling equipment.

Predicting levels of faecal contamination in shellfish

In the UK, the Food Standards Agency (FSA) classifies shellfish harvesting areas according to the concentration of *Escherichia coli* detected in shellfish flesh, following standards set out in assimilated EU regulations. The classification determines the level of treatment required before shellfish can be sold for human consumption. There are three main classes:

- **Class A**: At least 80% of samples must contain ≤230 *E. coli* per 100 g shellfish flesh, and no sample may exceed 700 *E. coli* per 100 g. Shellfish from Class A areas can be harvested for direct human consumption if end-product standards are met.
- **Class B**: At least 90% of samples must contain ≤4,600 *E. coli* per 100 g, and no sample may exceed 46,000 *E. coli* per 100 g. Shellfish from Class B areas require purification (relaying in a Class A area or heat treatment) before consumption.
- **Class C**: All samples must contain ≤46,000 *E. coli* per 100 g. Shellfish from Class C areas must be relayed for at least two months in an approved area or undergo heat treatment before they can be sold for human consumption.

If *E. coli* levels consistently exceed 46,000 per 100 g, the area is classified as 'Prohibited', and harvesting is not permitted. These classifications are based on regular microbiological monitoring, and the results directly inform public health protections and industry practices. Classification typically last for a year based on the microbiological data collected in the previous 12 months.

Modelling tools, such as the RDMAI classifier model, may be implemented to predict *E. coli* risk in water bodies of shellfish harvesting areas to assess water quality prior to site designation for shellfish harvesting. This could help provide actionable insights, such as flagging pollution sources and/or enabling the selection of non-contaminated areas for harvest. If data on the microbial quality of shellfish are available, the model may be further refined to predict harvesting area classification. To explore the use of *E. coli* predictive modelling relevant for the shellfish industry, two case studies are summarised below.

Case study 1: Short-term predictions of Escherichia coli levels in shellfish (Schmidt et al., 2018)

Summary

The study developed and evaluated models to predict short-term variations in *E. coli* concentrations in shellfish, using environmental data from two sites in the UK: St Austell Bay (coastal mussels harvesting area) and Turnaware Bar (estuarine oyster harvesting site). Rainfall (including antecedent rainfall), river flow, solar radiation, and sea surface temperature were identified as significant predictors for *E. coli* levels. These factors influenced contamination through mechanisms like runoff and

microbial die-off. The best-performing models predicted the European hygiene classification (equivalent of FSA regulations) of shellfish beds with 99% accuracy at St Austell Bay and 98% accuracy at Turnaware Bar. These models enable authorities and farmers to estimate likely *E. coli* variations in shellfish areas, even in the absence of regular monitoring. This supports decisions on when to harvest, when to avoid harvesting, and when to sell stock, thereby reducing health risks and improving management.

The study highlighted that modelling *E. coli* concentrations using accessible environmental data can provide a robust, high-accuracy tool for classifying shellfish beds and supporting farm management. Similar, generic approaches can be adapted to other sites and potentially other contaminants, making it highly relevant for sustainable aquaculture operations.

Schmidt et al., (2018) Workflow:

INPUTS
• Environmental data: rainfall, river flow, solar radiation, sea surface temperature (SST), wind speed, wind direction • Historical *E. coli* concentrations in shellfish (from routine monitoring) • Site-specific data: coastal bay (St Austell Bay) and estuary (Turnaware Bar) • Meteorological nowcasts and forecasts • Satellite observations • Regulatory thresholds for classification
PROCESSING/MODEL

• Statistical modelling using:
  ○ Generalized Linear Models (GLM)
  ○ Averaged GLMs (information-theoretic approach)
  ○ Generalized Additive Models (GAM)- Log transformation of response variables (*E. coli* concentrations)
• Model selection based on lowest Root Mean Square Error (RMSE)
• Identification of significant explanatory variables (rainfall, river flow, solar radiation, SST)
• Independent validation using confusion matrix analysis (accuracy, precision, false positive/discovery rates)
• Models trained to predict *E. coli* concentrations and classify shellfish beds (Class A/B)

OUTPUT
• Predicted short-term (1 week) *E. coli* concentrations in shellfish • Predicted hygiene classification of shellfish beds (Class A/B) • Model accuracy metrics (up to 99% for St Austell Bay, 98% for Turnaware Bar) • Decision support for farm management (harvesting, sales, risk mitigation)

• Early warning for potential closures due to contamination
END-USER INTERFACE / COMMUNICATION
• Results communicated to shellfish farmers and local authorities • Practical guidance for harvesting decisions (when to/not to harvest) • Support for regulatory compliance and consumer safety • Potential integration with farm management systems or dashboards • Recommendations for annual model updates to reflect changing environmental and farm conditions

Case study 2: Models predicting shellfish harvesting risk catchment to estuary (Malham et al., 2025)

Summary

This study evaluated predictors of *E. coli* in blue mussels and Pacific oysters across 12 estuaries in England and Wales. Modelling identified significant *E. coli* contamination predictors (river flow, turbidity and nitrate), while no effect from catchment area, number of point source discharges, or agricultural land use type was noted. *E. coli* counts were consistently higher when Storm Overflows (SOs) had been active in the previous week. A significant association between *E. coli* concentrations and river flow was observed in 64% of shellfish beds, with contamination events typically lagging river flow changes by 1 to 3 days. High variability among beds within the same estuary means that while estuary-level risk can be modelled, predicting risk at the individual bed level remained challenging.

Predictive models can help anticipate when beds are at risk of exceeding these thresholds. Real-time river flow and water quality data can be used to predict contamination risk, supporting more dynamic and responsive management (e.g., timely closures or re-openings). Models are more reliable at the estuary scale than at the individual shellfish bed scale due to local variability. Incorporating fine-scale hydrodynamic models and higher-frequency data could improve predictions for specific beds.

Malham et al., (2025) Workflow:

INPUTS
• Routine monthly *E. coli* measurements in shellfish flesh (2010–2017, 12 estuaries, 131 beds) • Environmental data: river flow, rainfall, turbidity, nitrate • Catchment characteristics: land use, catchment area • Point source data: number and location of sewage treatment works (STWs), Storm Overflows (SOs) • Spatial data: shellfish bed locations, estuary features
PROCESSING/MODEL
• Linear mixed effects models to identify significant predictors of *E. coli* levels • Lag regression analysis between environmental events (e.g., rainfall/river flow) and *E. coli* accumulation • Analysis of variance for SO events • Model validation using observed vs. predicted *E. coli* values
OUTPUT
• Predictive relationships between *E. coli* levels and environmental variables (river flow, turbidity, nitrate) • Identification of typical lag times (1–3 days) between river flow events and *E. coli* spikes • Estuary-level risk predictions for shellfish contamination • Evidence that SO operation is associated with higher *E. coli* in shellfish (case study)
END-USER INTERFACE / COMMUNICATION
• Recommendations for 'Active Management Systems' using real-time river flow and water quality data • Guidance for regulators and industry on when to close or reopen shellfish beds • Communication of risk to shellfish farmers, regulators, and potentially the public • Suggestion to integrate predictive models with real-time sensors and forecasting for dynamic management

Value Propositions

Predictive models for *E. coli* concentrations in shellfish harvesting areas offer several important benefits to stakeholders. The RDMAI classifier used as a forecasting model, with threshold adjustments for human consumption of shellfish and appropriate training data, could provide early warnings for deteriorating water and shellfish

quality that help protect both public health and industry revenue. By forecasting short-term contamination risks, these models can assist farmers and regulators to avoid harvesting during periods when closures are likely, and to adjust sales strategies ahead of risk windows. This proactive approach would reduce the likelihood of contaminated shellfish reaching consumers and helps businesses plan more effectively. Predictive modelling may also support the rapid, evidence-led reopening of shellfish beds. By accurately tracking whether contamination is accumulating or dissipating, these models allow for targeted resampling and earlier decisions to reopen beds. This improves operational efficiency and reduces unnecessary downtime.

The models can also provide confidence in bed classification. With the high accuracy rates noted for the specific models, predictive tools can justify operational choices to buyers and auditors, supporting regulatory compliance and market access. Additionally, catchment-to-bed risk flags generated by these models help explain more than half of the variation in contamination across estuaries. By identifying upstream risk factors such as river flow, nitrate, and turbidity, stakeholders could take action before contamination reaches shellfish beds.

The RDMAI *E. coli* classifier model can plausibly be modified to inform lag-aware harvest scheduling. By recognising typical delays of a few days between high river flow and mussel contamination, the model can inform the dynamic adjustment of harvest windows around storm events, further reducing risk and optimising production.

Limitations

This use-case would only be feasible with further training of the RDMAI *E. coli* model with additional datasets (i.e. shellfish flesh) and benchmarking against shellfish bed sites. The RDMAI *E. coli* classifier would also need to be trained on thresholds related to the safe human consumption of shellfish.

A major challenge in utilising predictive modelling for shellfish classification is the high degree of variability between individual shellfish beds, even within the same estuary. While catchment-scale predictors such as river flow, nitrate, and turbidity can be significant, the local differences across beds and tides are substantial. As a result, single predictive rules rarely transfer cleanly from one bed to another, making it difficult to generalise model outputs.

Additionally, the environmental drivers used in the models capture the likelihood of contamination events but do not provide detailed information about the actual contamination pathways, such as the timing, magnitude, and location of storm overflow pulses, livestock proximity, or slurry applications. This limits the operational specificity of the models. Another significant limitation is the frequency, granularity and uncertainty of available data. Regulatory sampling is typically conducted monthly, and environmental data series may be intermittent or lack real-time

telemetry. This creates blind spots during short-lived contamination peaks and recovery periods, which can extend closures or cause events to be missed altogether.

There is also a trade-off between model generalisability and local optimisation. Generic statistical frameworks such as GLM or GAM can be scaled easily, but tend to underperform compared to bespoke, bed-specific hydrobacterial models. However, these custom models require more instrumentation and periodic retuning, which can be resource intensive. The RDMAI *E. coli* classifier uses a flexible machine learning model that can be applied at different spatial and temporal scales.

Next steps and opportunities

In addition to improving RDMAI model accuracy, a dedicated case study is needed to refine the RDMAI model for shellfish harvesting applications. Applying the model specifically to predicting microbial water quality in shellfish harvesting areas and their catchments would help demonstrate the model's feasibility in these environments. Following that, the RDMAI model should be refined to incorporate *E. coli* data derived directly from shellfish. This enhancement would allow the model to more accurately predict the classification of harvesting areas, aligning outputs more closely with regulatory standards and industry needs.

The development of an Active Management System that integrates collected data and model outputs, enabling shellfish harvesting operations to respond dynamically to real-time environmental changes has been suggested. By leveraging predictive insights, such a system would support more responsive and sustainable management, helping to protect both public health and industry interests.

Supporting catchment risk assessment for drinking water abstraction

In the UK, risk assessment of drinking water abstraction points is a statutory requirement under The Water Supply (Water Quality) Regulations 2016 (as amended 2018) (England); The Water Supply (Water Quality) Regulations 2018 (Wales); The Public Water Supplies (Scotland) Regulations 2014; and The Water Supply (Water Quality) Regulations (Northern Ireland) 2017. Water suppliers must characterise hazards within the abstraction point catchment and demonstrate that control measures are in place to protect raw water quality.

Regulation 17 of the Water Supply (Water Quality) Regulations 2016 (England and Wales) requires routine monitoring of abstraction points, including microbiological parameters, to inform the risk-based approach to water treatment set out in Regulations 26-28, which mandate adequate disinfection and ongoing risk assessment of treatment and supply. Together, these regulations support compliance with Regulation 4, which requires that drinking water supplied to consumers is wholesome and safe to drink.

E. coli is a prescribed indicator organism under these regulations and provides evidence of faecal contamination in water sources. Routine *E. coli* monitoring must be conducted at all abstraction points to support ongoing catchment risk assessments. While no regulatory compliance thresholds are specified for *E. coli* in raw water, water suppliers are expected to establish their own internal thresholds or trigger levels for *E. coli* as part of their risk assessments to support operational decisions including enhanced monitoring, catchment controls, or treatment optimisation, to ensure that treated water remains wholesome.

The RDMAI *E. coli* classifier model could be used to support water suppliers in catchment risk assessments by providing real time or forecasted information on *E. coli* risk levels in source waters. The model could be tailored to specific abstraction catchments by inputting site-specific microbiological, land use and meteorological data, which are routinely held by water company catchment teams. This would help suppliers to identify periods of heightened microbiological risk and inform operational decision-making at the abstraction point.

The following section demonstrates two case studies where microbial water quality has been simulated using predictive models. These case studies are included to illustrate how *E. coli* forecasting models can be applied to abstraction point characterisation. It should be noted, however, that the RDMAI *E. coli* model is not intended as a direct replacement for the approaches described and would require substantial modification to achieve comparable functionality, particularly with regards to Case Study 1.

Case study 1: Forecasting acute rainfall driven *E. coli* impacts in inland rivers based on sewer monitoring and field runoff (Suslovaite et al., 2024)

Summary

The model developed in this study was designed to forecast short-term changes in *E. coli* concentrations at a drinking water abstraction site in the River Leam, a 300-km^2 sub-catchment of the River Severn, UK. The model combines meteorological data, travel-time based runoff routing, land-used derived diffuse faecal loading, and real time sewer overflow monitoring to simulate how faecal contamination moves from agricultural land and storm sewer systems to a river monitoring point.

The model accounts for two major faecal pathways relevant to drinking water abstraction:

- Diffuse runoff from agricultural land mobilised during rainfall events was represented using livestock-derived *E. coli* loadings, die-off rates and rainfall-driven runoff
- Point source storm sewer overflow discharges were represented using spill volumes and timings

Hydrological travel-time routing was used to simulate how loads from different parts of the catchment reach the river. High-resolution radar rainfall data was used to inform the runoff model. The combined diffuse and point-source loadings were routed to the monitoring point to generate predicted *E. coli* time series.

The model was calibrated using one rainfall event and validated against three additional events across different seasons. Across validation events, the model successfully reproduced the timing and duration of high *E. coli* concentrations, showing that peak contamination concentrations could be forecasted hours in advance. Notably the modelling demonstrated that contamination sources varied based on event characteristics; intense rainfall was associated mostly with storm sewer overflow contributions while longer events showed greater diffuse loading from agricultural land. Understanding how different rainfall patterns drive changes in source contributions is valuable for catchment risk assessment as it helps water suppliers identify which parts of the catchment pose the greatest risk under specific weather conditions, and target monitoring and mitigation measures accordingly.

Suslovaite et al., (2024) Workflow:

INPUTS
Catchment data • Rainfall radar data (1-km^2 spatial resolution, 15-min temporal resolution) • River flow (m^3/s) • Land cover and grazing land distribution • Livestock densities and associated daily *E. coli* excretion rates, taking account of slurry spreading • Bacterial die-off rates applied to faecal matter on grazing land • Digital elevation model and hydrological flow paths • Storm sewer overflow discharge volumes and timing from automated spill monitors Microbiological data • High frequency *E. coli* measurements for calibration and validation • Routine abstraction point *E. coli* monitoring
PROCESSING/MODEL
• Surface runoff modelling: calculated based on overland flow from 5-m^2 grid cells in catchment, using radar rainfall data. GIS used to estimate storm runoff travel time from catchment grid cells to outlet of catchment. This method captures how runoff varies across the catchment spatially and temporally, based on hourly estimates. • Diffuse *E. coli* loading modelling: estimated based on livestock densities and land use, and surface runoff into water course. • Point source storm sewer overflow modelling: volume and loading from storm sewer overflows calculated based on level data from storm overflow monitors, which is provided at 15-min resolution. • Diffuse and point source impacts are then combined to enable prediction of *E. coli* concentrations after rainfall events.
OUTPUT
• Predicted *E. coli* concentration time series at the river monitoring point • Forecasted arrival time, magnitude, and duration of *E. coli* peaks following rainfall • Identification of event-specific dominant contamination sources (diffuse vs point source)
END-USER INTERFACE / COMMUNICATION
• For internal use by water suppliers as opposed to public messaging/alerts • Generation of short-term forecasts to provide advance warning of water quality risks at abstraction site

• Integration within existing risk assessment and water safety planning frameworks by water suppliers

Case study 2: Data-driven models for predicting microbial water quality in the drinking water source using *E. coli* monitoring and hydrometeorological data (Sokolova et al., 2022)

Summary

This study developed data-driven forecasting models to predict changes in *E. coli* concentrations in the river Göta älv at the intake for a drinking water treatment plant serving a population of 500,000 in Gothenburg, Sweden. Approaches of varying complexity were explored to optimise performance for forecasting of *E. coli* one day ahead. These ranged from traditional univariate time-series approaches to multivariate machine learning methods. Predictor variables included historical *E. coli* data (laboratory and Colifast β-galactosidase sensor measurements), total coliforms, water temperature, turbidity, precipitation, and river flow.

Among the tested approaches, multivariate models that integrated multiple predictors (e.g. Random Forest, VAR, LASSO Regression, TPOT) outperformed univariate baselines (e.g. naive forecasts, Exponential Smoothing, ARIMA). Machine learning methods, specifically Random Forest and TPOT (Tree-based Pipeline Optimization Tool), demonstrated the highest predictive performance, though there were indications of overfitting in some cases. Key predictors were upstream precipitation, microbial concentrations, and water temperature.

The authors highlight that combining data-driven *E. coli* forecasting models with rapid, online microbial sensing provides a promising approach for near-real-time characterisation of faecal contamination at drinking water sources.

Sokolova et al., (2022) Workflow:

INPUTS
Catchment data • Precipitation (daily sum) • Water temperature and turbidity at the intake (average each 2–3 days) • River flow (daily mean) Microbiological data • Laboratory *E. coli* and coliform measurements (average each 2–3 days) • Colifast at-line β-galactosidase *E. coli* sensor measurements (twice daily)

PROCESSING/MODEL
Multiple models explored including: Univariate modelling • Naïve forecast • Exponential Smoothing • ARIMA (Autoregressive Integrated Moving Average) Multivariate modelling • VAR (Vector Autoregression) • LASSO ((Least Absolute Shrinkage and Selection Operator) Regression • Random Forest • TPOT (Tree-based Pipeline Optimization Tool) Best performance: Random Forest, TPOT Best balance between performance and generalization: LASSO Regression, VAR
OUTPUT
• Forecasted *E. coli* concentrations one day ahead for the Göta älv drinking water intake.
END-USER INTERFACE / COMMUNICATION
• The modelling framework is intended to support operational decision-making by water suppliers rather than public alerting, providing short-term probabilistic forecasts of microbial contamination risk. • Supports integration with automated monitoring systems e.g. Colifast

Value propositions

The benefit of using an *E. coli* model such as the RDMAI *E. coli* classifier model to support catchment risk assessment is that it offers an automated and rapid method of characterising the microbial water quality of an entire river catchment. As demonstrated in the above case studies, this is particularly valuable when forecasting the impacts of rainfall events, which directly impact water quality via the introduction of rainfall-driven diffuse and point sources of faecal contamination to a waterbody. While the RDMAI *E. coli* model does not account for specific faecal sources, as described in Case Study 1 (see Limitations), Case Study 2 demonstrates that more general water quality parameters (e.g. precipitation, water temperature, turbidity) can be used to forecast *E. coli* concentrations in drinking water sources. This supports the application of the RDMAI model for abstraction point risk assessment purposes.

Model outputs could be used to inform sampling location and schedule by identifying time periods and areas within the catchment associated with elevated predicted *E. coli* risk levels, thereby enabling targeted monitoring. Such use would be

compliant with regulatory requirements provided it does not displace mandatory minimum sampling frequencies or any requirements specified in Regulation 17 notices and is implemented within a clearly governed risk-based monitoring framework.

For this use case, water suppliers represent the primary end users and are likely to have access to catchment-specific datasets, including routine regulatory *E. coli* measurements, land-use information, and hydrometeorological data, and *E. coli* sensor data. Incorporation of these datasets would improve model performance and relevance at the abstraction scale. Predictive modelling could be integrated into existing water company water safety planning frameworks as a complementary tool to identify high-risk areas of the catchment, to identify higher-risk periods within the catchment, and to inform targeted monitoring and control strategies alongside conventional field- and laboratory-based testing.

Limitations

Routine *E. coli* monitoring at the abstraction point is a regulatory requirement in the UK. Consequently, the RDMAI *E. coli* model would not be considered as a replacement for traditional monitoring, but rather as a complementary tool to inform risk assessments. Given the limitations of the model regarding accuracy and false negatives as mentioned previously, the tool should be viewed by water suppliers as a high-level forecast of potential *E. coli* peaks as opposed to a definitive indicator of water quality. Any use of model outputs to inform changes to disinfection practices or catchment management would require appropriate justification and supporting evidence to satisfy regulatory expectations.

To use the model to its fullest potential for this catchment risk assessment, it would likely need to be tailored and calibrated for individual catchments, which would require additional time and resource investment from water suppliers. Additionally, the RDMAI model, in its current form, does not account for point and diffuse faecal sources, such as storm sewer overflows or livestock inputs, as the Case Study 1 model does. As a result, its ability to provide direct source attribution is limited, although broad inferences may be drawn using land-use data, water quality data and supporting contextual information. Since the RDMAI models are open source, there is potential to re-train them, incorporating data on direct sources of contamination.

It should also be recognised that UK abstraction point monitoring covers a wide range of parameters, and *E. coli* alone cannot act as a proxy for all regulated water quality risks, particularly those unrelated to faecal contamination (e.g. chemical indicators). Predictive modelling should therefore be considered a targeted tool supporting one component of catchment characterisation, rather than a comprehensive solution for abstraction risk assessment.

Next steps and opportunities

This use case illustrates how *E. coli* models such as the RDMAI *E. coli* classifier models could be applied to support catchment risk assessment for drinking water abstraction by providing an early indication of microbiological risk during high rainfall events. Water suppliers are required to assess risks to abstraction points, implement appropriate control measures, and ensure that treatment processes are sufficient to deliver wholesome water. Predictive *E. coli* models could support this by informing where and when microbiological risks are likely to arise within a catchment. The ability to provide more spatial and temporal resolution than routine laboratory sampling alone is particularly valuable during rainfall events, which may be short-lived and difficult to characterise using conventional monitoring approaches.

There is opportunity for integration within existing water safety planning frameworks, where model outputs could be aligned with internal *E. coli* thresholds to provide an early warning of potential exceedances. Additionally, there is opportunity to combine modelling approaches with emerging rapid/automated *E. coli* monitoring techniques.

Conclusions

The RDMAI classifier model for predicting *Escherichia coli* risk levels riverine and coastal UK waters has potential to enhance water quality management across diverse applications, including bathing water safety, shellfish harvesting, and drinking water abstraction risk assessment. By leveraging environmental, meteorological, and water quality data, the model provides early warnings of faecal contamination risks, enabling proactive decision-making that supports public health protection, regulatory compliance, and operational efficiency.

While the model offers advantages over traditional monitoring, such as improved spatial and temporal resolution and potential integration with digital communication platforms, it should be regarded as a complementary tool rather than a replacement for statutory sampling. Limitations related to data availability, calibration requirements, and predictive uncertainty underscore the need for careful interpretation and ongoing validation.

Future opportunities lie in scaling the model to broader geographic areas, refining algorithms with additional data sources (e.g., shellfish-derived *E. coli*), and embedding predictive outputs within policy frameworks and industry workflows. These steps will strengthen its role in safeguarding water resources and ensuring sustainable, evidence-based management practices.

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